

Ning Liang-bo



Homepage



Google Scholar (Citations: 2,041)



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Education

- Jan. 2026 – Present **Visiting Ph.D. Student, *Computing and Data Science***
Nanyang Technological University (NTU), Singapore
Supervisor: Prof. Dacheng Tao
- Sep. 2023 – Present **Ph.D., *Computing*, GPA:3.49/4.3**
The Hong Kong Polytechnic University (PolyU), Hong Kong, China
Supervisor: Prof. Qing Li & Prof. Wenqi Fan
- Sep. 2020 – Jun. 2023 **M.E., *Control Science and Engineering*, GPA:88.00/100 or 3.75/4**
Northwestern Polytechnical University (NPU), Xi'an, China
Supervisor: Prof. Zhun-ga Liu
- Sep. 2016 – Jun. 2020 **B.E., *Automation*, GPA:83.47/100 or 3.34/4**
Northwestern Polytechnical University (NPU), Xi'an, China

Publications (First Author)

- 2026 **L. Ning**, Y. Zhu, H. Huang, et al., “When Efficiency Becomes a Vulnerability: Computational Cost Attacks on WebAgents.” *Annual Meeting of the Association for Computational Linguistics (ACL '26 Main)*, 2026. (CCF-A)
- L. Ning**, W. Fan, Q. Li, “Exploring Backdoor Attack and Defense for LLM-empowered Recommendations.” *arXiv*, 2026. (**Preprint, TKDE Major Revision**)
- L. Ning**, Z. Dai, W. Fan, et al., “Joint Universal Adversarial Perturbations with Interpretations,” *Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD '26)*, 2024. (CCF-C)
- C. Liu*, **L. Ning***, W. Fan, Q. Li, “Inference Cost Attacks for Retrieval-Augmented Large Language Models.” *ACM WEB Conference 2026 (WWW '26)*, 2026. (CCF-A)
- L. Ning**, W. Fan, Q. Li, “Retrieval-Augmented Purifier for Robust LLM-Empowered Recommendation.” *Transactions on Information Systems (TOIS)*, 2026. (CCF-A)
- 2025 **L. Ning**, Z. Liang, Z. Jiang, et al., “A Survey of WebAgents: Towards Next-Generation AI Agents for Web Automation with Large Foundation Models.” *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '25)*, 2025. (CCF-A)
- L. Ning**, Z. Zhang, W. Ding, et al., “Multi-level Distribution Alignment for Multi-source Universal Domain Adaptation.” *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, 2025. (JCR:Q1, IF:10.2) [ESI Highly Cited Paper (2025 Nov / 2026 Jan.)]
- 2024 **L. Ning**, S. Wang, W. Fan, et al., “CheatAgent: Attacking LLM-Empowered Recommender Systems via LLM Agent,” *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24)*, 2024. (CCF-A)
- L. Ning**, Z. Dai, J. Su, et al., “Interpretation-Empowered Neural Cleanse for Backdoor Attacks,” *ACM Web Conference (WWW '24)*, 2024. (CCF-A, Short Paper)
- 2022 Z. Liu (**Master Supervisor**), **L. Ning**, and Z. Zhang, “A New Progressive Multi-Source Domain Adaptation Network with Weighted Decision Fusion,” *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, 2022. (JCR:Q1, IF:10.2)
- 2021 **L. Ning**, Z. Liu, and Z. Zhang, “Domain Adaptation Based Multi-Source Data Fusion for Pattern Classification,” *CIE International Conference on Radar (Radar)*, 2021.

Publications (Collaboration)

- 2026  J. Huang, S. Wang, **L. Ning**, W. Fan, Q. Li, "ReRec: Reasoning-Augmented LLM-based Recommendation Assistant via Reinforcement Fine-tuning." *Annual Meeting of the Association for Computational Linguistics (ACL '26 Main)*, 2026. (CCF-A)
-  X. Yuan, **L. Ning**, W. Fan, and Q. Li. "mKG-RAG: Multimodal Knowledge Graph-Enhanced RAG for Visual Question Answering." *ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '26)*, 2026. (CCF-A)
-  H. Qu, **L. Ning**, R. An, W. Fan, et al., "A Survey of Mamba," *ACM Transactions on Intelligent Systems and Technology (TIST)*, 2024. (JCR:Q1, IF:8.4)
- 2025  Y. Zhang, H. Qu, **L. Ning**, et al. "SSD4Rec: a structured state space duality model for efficient sequential recommendation." *Transactions on Information Systems (TOIS)*, 2025. (CCF-A)
-  Z. Zhang, Z. Liu, **L. Ning**, et al. "Belief-based fuzzy and imprecise clustering for arbitrary data distributions." *IEEE Transactions on Fuzzy Systems (TFS)*, 2025. (JCR:Q1, IF:10.8)
-  J. Huang, S. Wang, **L. Ning**, et al., "Towards Next-Generation Recommender Systems: A Benchmark for Personalized Recommendation Assistant with LLMs." *ACM International Conference on Web Search and Data Mining (WSDM '25)*, 2025. (CCF B)
- 2024  Y. Ding, W. Fan, **L. Ning**, et al., "A survey on RAG meets LLMs: Towards retrieval-augmented large language models," *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24)*, 2024. (CCF-A)
- 2023  Z. Zhang, Z. Liu, **L. Ning**, et al., "Representation of Imprecision in Deep Neural Networks for Image Classification," *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, 2023. (JCR:Q1, IF:10.2)
-  Z. Zhang, **L. Ning**, Z. Liu, et al., "Mining and Reasoning of Data Uncertainty-induced Imprecision in Deep Image Classification," *Information Fusion*, 2023. (JCR:Q1, IF:10.2)
- 2022  Z. Liu, Z. Zhang, Q. Pan, and **L. Ning**, "Unsupervised Change Detection From Heterogeneous Data Based on Image Translation," *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*, vol. 60, pp. 1-13, 2022. (JCR:Q1, IF:8.125)

Tutorial

- 2025  **L. Ning**, Z. Liang, Z. Jiang, et al., "WebAgents: Towards Next-Generation AI Agents for Web Automation with Large Foundation Models" *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '25)*, 2025.
-  P. Wu, Y. Ding, **L. Ning**, W. Fan, et al., "Towards Retrieval-Augmented Large Language Models: Data Management and System Design," *IEEE International Conference on Data Engineering (ICDE '25)*, 2025.
- 2024  Y. Ding, W. Fan, **L. Ning**, et al., "RAG Meets LLMs: Towards Retrieval-Augmented Large Language Models," *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24)*, 2024.
-  Y. Ding, W. Fan, **L. Ning**, et al., "Recommender Systems in the Era of Large Language Models (LLMs)," *International Joint Conference on Artificial Intelligence (IJCAI '24)*, 2024.

Academic Services

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|-----------------------|--|
| Conference Area Chair |  CIKM'25 - MMGenSR Workshop, IEEE ATC 2024 |
| Conference Reviewing |  KDD - 24/25/26, WWW - 24/25, IJCAI - 24, ACMMM - 25, ACL - 25/26 |
| Journal Reviewing |  IEEE TKDE, IEEE TNNLS, IEEE TSMC, ACM TOIS, ACM TIST, Pattern Recognition, Knowledge-based Systems, Neurocomputing |

Miscellaneous Experience

Awards and Achievements

- 2024  **KDD 2025 Commendable Reviewers**, SIGKDD.
-  **KDD 2024 Student Travel Award**, SIGKDD.
- 2023  **Excellent Master's Thesis**, NPU, Xi'an, China.
-  **Outstanding Graduates**, NPU, Xi'an, China.
- 2022  **The National Scholarship**, Ministry of Education, China.
-  **The First Prize Scholarship**, NPU, Xi'an, China.
- 2019  **The Second Prize Scholarship**, NPU, Xi'an, China.
- 2018  **The First Prize Scholarship**, NPU, Xi'an, China.
-  **The Endress+Hauser Scholarship**, NPU, Xi'an, China.
- 2017  **The Second Prize Scholarship**, NPU, Xi'an, China.

Certifications

- 2020  **The First Prize of the 15-th National University Students Intelligent Car Race.**
- 2019  **The First Prize of the 14-th National University Students Intelligent Car Race.**
- 2018  **The Second Prize of the 13-th National University Students Intelligent Car Race.**
Awarded by Intelligent Car Race Organization Committee.